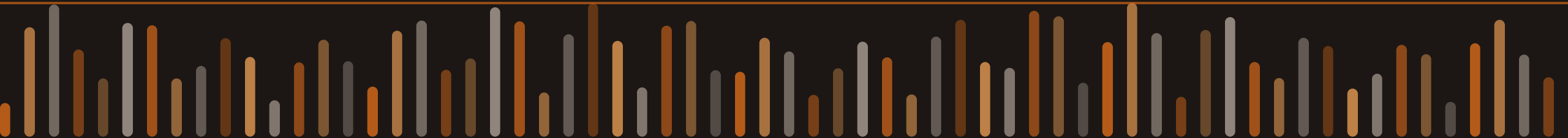


DATA CLEANING CASE STUDY · POWER QUERY IN EXCEL

Audible Data Cleaning Case Study

Cleaning, standardizing, and restructuring a raw Audible audiobook dataset into an analysis-ready format.

Presented by Samarth Pathak



PROJECT OVERVIEW & OBJECTIVES

From Raw Export to Analysis-Ready Data

The raw Audible export arrived with inconsistent title casing, combined author and narrator names, text-based durations and star ratings, non-numeric price entries, and mismatched date formats.

This case study uses Power Query in Excel to diagnose and resolve nine distinct data-quality issues — turning the dataset into a clean, consistent, and reliable foundation for analysis.

TOOL

Microsoft Excel — Power Query Editor

Used for column transformation, type conversion, delimiter splitting, error handling, and custom-column formulas (M language).

CASE STUDY ROADMAP • 9 QUESTIONS

Text Standardization

- 1 Standardize NAME casing
- 2 Split combined AUTHOR names

Date & Duration Formatting

- 3 Format RELEASEDATE consistently
- 4 Convert TIME to true duration

Numeric Data Cleanup

- 5 Clean & validate PRICE values
- 6 Convert STARS text to numeric

Column Restructuring

- 7 Split NARRATEDBY into columns
- 8 Merge date + language → RELEASEINFO

Currency Precision

- 9 Format PRICE to 2 decimal places

SECTION • Text Standardization

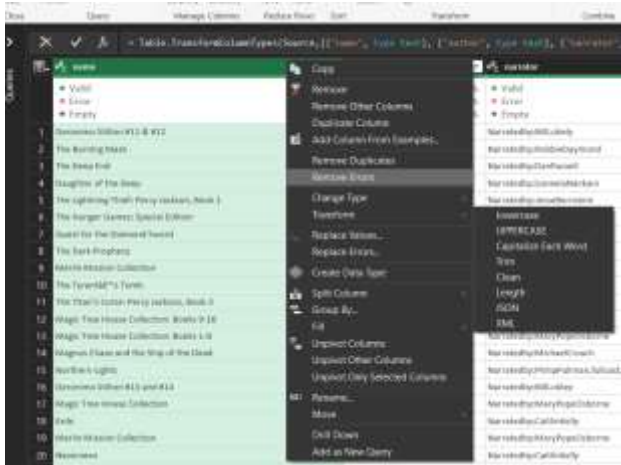
QUESTION 1 OF 9

Standardize the NAME column for consistent title casing

TRANSFORMATION

Applied “Capitalize Each Word” under the Transform menu to normalize inconsistent book-title casing across all 981 distinct entries in the name column.

Applying the transform



Result — consistent title casing

	name	author	narrator
	Valid Error Empty	Valid Error Empty	Valid Error Empty
1	Geronimo Stilton #11 & #12	Writtenby:GeronimoStilton	Narratedby:BillLobely
2	The Burning Maze	Writtenby:RickRiordan	Narratedby:RobbieDaymond
3	The Deep End	Writtenby:JeffKinney	Narratedby:DanRussell
4	Daughter Of The Deep	Writtenby:RickRiordan	Narratedby:SoneelaNankani
5	The Lightning Thief: Percy Jackson, Book 1	Writtenby:RickRiordan	Narratedby:JesseBernstein
6	The Hunger Games: Special Edition	Writtenby:SuzanneCollins	Narratedby:TatianaMaslany
7	Quest For The Diamond Sword	Writtenby:WinterMorgan	Narratedby:LukeDaniels
8	The Dark Prophecy	Writtenby:RickRiordan	Narratedby:RobbieDaymond

SECTION • Text Standardization

QUESTION 2 OF 9

Separate combined names in the AUTHOR column

TRANSFORMATION

Removed the “Writtenby:” prefix using Replace Values, then split the column by comma delimiter — isolating multiple co-authors into their own fields.

Step 1 — Remove prefix



Step 2 — Split by delimiter



Result — author.1 through author.4

author.1	author.2	author.3	author.4	narrator
Valid 100% Error 0% Empty 0%	Valid 21% Error 0% Empty 79%	Valid 2% Error 0% Empty 98%	Valid 0% Error 0% Empty 100%	Valid 0% Error 0% Empty 100%
RobinStevens				Narratedby:KatieLeung_Rebecca
DavidWalliams				Narratedby:DavidWalliams_Niti
CarolynKeene				Narratedby:JorjeanaMarie
M.G.Leonard	SamSedgman	ElisaPaganelli-Illustrator		Narratedby:JotDavies
ShannonMessenger				Narratedby:CastlinKelly
TomFletcher				Narratedby:TomFelton,TomFle
TerriFarley				Narratedby:NatalieBudig
ShannonHale	DeanHale			Narratedby:JuliaWhelan
CarolynKeene				Narratedby:JorjeanaMarie
TraceyWest				Narratedby:EmilyEllet
FranklinW.Dixon				Narratedby:TimGregory

Ensure RELEASEDATE follows a consistent DD-MM-YYYY format

TRANSFORMATION

Changed the column's data type to Date, letting Power Query enforce one uniform DD-MM-YYYY format across all 410 distinct release dates.

Result — releasedate as Date type

releasedate	language	stars	price
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid -% Error < 1% Empty -%
04-08-2008	English	5 out of 5 stars34 ratings	4
01-05-2018	English	4.5 out of 5 stars41 ratings	8
06-11-2020	English	4.5 out of 5 stars38 ratings	4
05-10-2021	English	4.5 out of 5 stars12 ratings	6
13-01-2010	English	4.5 out of 5 stars181 ratings	8
30-10-2018	English	5 out of 5 stars72 ratings	6
25-11-2014	English	5 out of 5 stars11 ratings	2
02-05-2017	English	5 out of 5 stars50 ratings	6
02-05-2017	English	5 out of 5 stars5 ratings	12
24-09-2019	English	5 out of 5 stars58 ratings	8
14-01-2010	English	4.5 out of 5 stars130 ratings	8
24-08-2011	English	5 out of 5 stars6 ratings	12
27-09-2011	English	5 out of 5 stars7 ratings	12

Convert TIME from text to a duration format Excel recognizes

TRANSFORMATION

Replaced “hrs and” / “hr and” with “:” via Replace Values, then added a custom column with conditional logic to compute a true Duration value.

04

Step 1 — Replace text separators

time	releasedate	language	stars
2 hrs and 20 mins	04-08-2008	English	5 out of 5 stars34 ratings

Replace Values

Replace one value with another in the selected columns.

Value To Find

hrs and

Replace With

:

Advanced options

OK

Cancel

Step 2 — Custom column formula

Custom Column

Add a column that is computed from other columns.

New column name: Duration

Custom column formula:

```
= IF([time] <> "",  
  IF(LEFT([time], 2) = "hrs",  
    IF(LEFT([time], 1) = "h",  
      VALUE([time]) * 60 + VALUE(LEFT([time], 10)),  
      VALUE([time]) * 60 + VALUE(LEFT([time], 10))),  
  0)
```

OK

Cancel

QUESTION 5 OF 9

TRANSFORMATION

Changed price to Decimal Number, duplicated the column, and used Keep Errors to isolate non-numeric entries (such as "Free") before resolving them to null.

ABC price		1.2 price - Copy	
100%	Valid	100%	Valid
0%	Error	0%	Error
0%	Empty	0%	Empty
	Free	Error	- %
gs	Free	Error	100%
	Free	Error	- %
	Free	Error	
	Free	Error	
	Free	Error	
	Free	Error	
	Free	Error	
	Free	Error	
	Free	Error	
	Free	Error	
	Free	Error	
	Free	Error	
gs	Free	Error	
	Free	Error	
	Free	Error	
gs	Free	Error	
gs	Free	Error	

update	language	stars	price
100%	Valid	100%	Valid
0%	Error	0%	Error
0%	Empty	0%	Empty
04-08-2003	English	3 out of 5 stars34 ratings	489
01-05-2010	English	4.5 out of 5 stars41 ratings	820
06-11-2010	English	4.5 out of 5 stars38 ratings	430
09-10-2011	English	4.5 out of 5 stars32 ratings	623
12-01-2010	English	4.5 out of 5 stars31 ratings	820
30-10-2010	English	5 out of 5 stars72 ratings	859
25-11-2014	English	3 out of 5 stars31 ratings	243
02-05-2017	English	5 out of 5 stars10 ratings	820
02-05-2017	English	5 out of 5 stars5 ratings	1238
04-09-2019	English	5 out of 5 stars58 ratings	820
14-03-2010	English	4.5 out of 5 stars330 ratings	820
24-08-2011	English	5 out of 5 stars6 ratings	1208
27-09-2011	English	5 out of 5 stars7 ratings	1206
03-10-2017	English	5 out of 5 stars4 ratings	820
14-06-2021	English	4 out of 5 stars2 ratings	1094
08-02-2020	English	4.5 out of 5 stars12 ratings	487
26-12-2004	English	5 out of 5 stars7 ratings	1206
06-11-2018	English	5 out of 5 stars20 ratings	836

SECTION • Numeric Data Cleanup

QUESTION 6 OF 9

Convert text ratings in the STARS column to numeric values

TRANSFORMATION

Extracted the text before the delimiter “out of 5 stars” and converted it to Decimal Number, then replaced any remaining errors with null.

Step 1 — Extract & convert to decimal

release date	language	1.2 stars	1.2 price
04-08-2008	English	4.5 out of 5 stars	4
01-09-2018	English	4.3 out of 5 stars	5
08-11-2020	English	4.3 out of 5 stars	6
05-10-2021	English	4.3 out of 5 stars	7
11-01-2019	English	4.4 out of 5 stars	8
10-10-2018	English	4.5 out of 5 stars	9
25-11-2014	English	4.5 out of 5 stars	10
02-03-2017	English	4.5 out of 5 stars	11
02-08-2017	English	4.5 out of 5 stars	12
24-09-2019	English	4.5 out of 5 stars	13
14-01-2010	English	4.5 out of 5 stars	14
24-08-2011	English	4.5 out of 5 stars	15
27-09-2011	English	4.5 out of 5 stars	16
01-10-2017	English	4.5 out of 5 stars	17
24-05-2021	English	4.5 out of 5 stars	18
08-02-2008	English	4.5 out of 5 stars	19
26-12-2004	English	4.5 out of 5 stars	20
05-11-2016	English	4.5 out of 5 stars	21
02-03-2017	English	4.5 out of 5 stars	22

Step 2 — Replace errors with null

language	1.2 stars	1.2 price
English	4.5 out of 5 stars	4
English	4.3 out of 5 stars	5
English	4.3 out of 5 stars	6
English	4.3 out of 5 stars	7
English	4.4 out of 5 stars	8
English	4.5 out of 5 stars	9
English	4.5 out of 5 stars	10
English	4.5 out of 5 stars	11
English	4.5 out of 5 stars	12
English	4.5 out of 5 stars	13
English	4.5 out of 5 stars	14
English	4.5 out of 5 stars	15
English	4.5 out of 5 stars	16
English	4.5 out of 5 stars	17
English	4.5 out of 5 stars	18
English	4.5 out of 5 stars	19
English	4.5 out of 5 stars	20
English	4.5 out of 5 stars	21
English	4.5 out of 5 stars	22
English	4.5 out of 5 stars	23
English	4.5 out of 5 stars	24
English	4.5 out of 5 stars	25
English	4.5 out of 5 stars	26
English	4.5 out of 5 stars	27
English	4.5 out of 5 stars	28
English	4.5 out of 5 stars	29
English	4.5 out of 5 stars	30
English	4.5 out of 5 stars	31
English	4.5 out of 5 stars	32
English	4.5 out of 5 stars	33
English	4.5 out of 5 stars	34
English	4.5 out of 5 stars	35
English	4.5 out of 5 stars	36
English	4.5 out of 5 stars	37
English	4.5 out of 5 stars	38
English	4.5 out of 5 stars	39
English	4.5 out of 5 stars	40
English	4.5 out of 5 stars	41
English	4.5 out of 5 stars	42
English	4.5 out of 5 stars	43
English	4.5 out of 5 stars	44
English	4.5 out of 5 stars	45
English	4.5 out of 5 stars	46
English	4.5 out of 5 stars	47
English	4.5 out of 5 stars	48
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English	4.5 out of 5 stars	70
English	4.5 out of 5 stars	71
English	4.5 out of 5 stars	72
English	4.5 out of 5 stars	73
English	4.5 out of 5 stars	74
English	4.5 out of 5 stars	75
English	4.5 out of 5 stars	76
English	4.5 out of 5 stars	77
English	4.5 out of 5 stars	78
English	4.5 out of 5 stars	79
English	4.5 out of 5 stars	80
English	4.5 out of 5 stars	81
English	4.5 out of 5 stars	82
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English	4.5 out of 5 stars	84
English	4.5 out of 5 stars	85
English	4.5 out of 5 stars	86
English	4.5 out of 5 stars	87
English	4.5 out of 5 stars	88
English	4.5 out of 5 stars	89
English	4.5 out of 5 stars	90
English	4.5 out of 5 stars	91
English	4.5 out of 5 stars	92
English	4.5 out of 5 stars	93
English	4.5 out of 5 stars	94
English	4.5 out of 5 stars	95
English	4.5 out of 5 stars	96
English	4.5 out of 5 stars	97
English	4.5 out of 5 stars	98
English	4.5 out of 5 stars	99
English	4.5 out of 5 stars	100

SECTION · Column Restructuring

QUESTION 7 OF 9

Split NARRATEDBY into multiple columns for multiple narrators

TRANSFORMATION

Split the narratedby column by comma delimiter and removed the resulting empty trailing column — exposing narrator.1, narrator.2, and narrator.3.

Result — narrator.1 / .2 / .3

narrator.1			narrator.2			narrator.3			time	Duration	refreshdate
1%	Valid	100%	1%	Valid	100%	1%	Valid	100%	1%	Valid	100%
0%	Error	0%	0%	Error	0%	0%	Error	0%	0%	Error	0%
0%	Empty	0%	0%	Empty	0%	0%	Empty	0%	0%	Empty	0%
null	NarratedBy:BillLobby			null	null			null	2 hrs and 30 mins	0:02:30:00	
null	NarratedBy:RobbieDaymond			null	null			null	12 hrs and 8 mins	0:12:08:00	
null	NarratedBy:CarliKusell			null	null			null	2 hrs and 3 mins	0:02:03:00	
null	NarratedBy:SonelLafantkani			null	null			null	11 hrs and 10 mins	0:11:10:00	
null	NarratedBy:JesseBerrstein			null	null			null	10 hrs	0:10:00:00	
null	NarratedBy:TatianaMasiary			null	null			null	10 hrs and 35 mins	0:10:35:00	
null	NarratedBy:LukeDaniels			null	null			null	2 hrs and 35 mins	0:02:35:00	
null	NarratedBy:RobbieDaymond			null	null			null	12 hrs and 32 mins	0:12:32:00	
null	NarratedBy:MaryPopeOsborne			null	null			null	10 hrs and 58 mins	0:10:58:00	
null	NarratedBy:RobbieDaymond			null	null			null	12 hrs and 23 mins	0:12:23:00	
null	NarratedBy:JesseBerrstein			null	null			null	5 hrs and 48 mins	0:05:48:00	
null	NarratedBy:MaryPopeOsborne			null	null			null	5 hrs and 26 mins	0:05:26:00	
null	NarratedBy:MaryPopeOsborne			null	null			null	6 hrs and 1 min	0:06:01:00	
null	NarratedBy:MichaelCrosby			null	null			null	12 hrs and 58 mins	0:12:58:00	
null	NarratedBy:PhilipPullman			Fullcast	RuthWilliams			null	11 hrs and 55 mins	0:11:55:00	
null	NarratedBy:BillLobby			null	null			null	2 hrs and 25 mins	0:02:25:00	
null	NarratedBy:MaryPopeOsborne			null	null			null	5 hrs and 4 mins	0:05:04:00	
null	NarratedBy:CarliKusell			null	null			null	14 hrs and 41 mins	0:14:41:00	
null	NarratedBy:MaryPopeOsborne			null	null			null	10 hrs and 19 mins	0:10:19:00	
null	NarratedBy:CarliKusell			null	null			null	16 hrs and 42 mins	0:16:42:00	
null	NarratedBy:RobbieDaymond			null	null			null	12 hrs and 12 mins	0:12:12:00	
null	NarratedBy:GemmaDoyle			null	null			null	23 hrs and 29 mins	0:23:29:00	
null	NarratedBy:GerryO'Brien			null	null			null	6 hrs and 46 mins	0:06:46:00	

SECTION · Column Restructuring

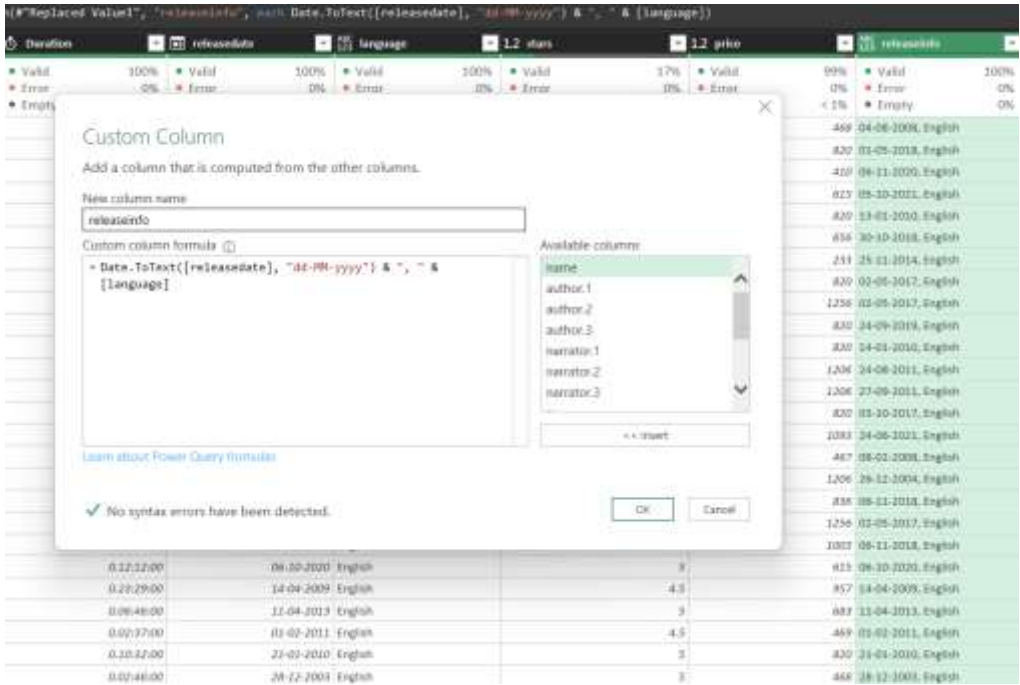
QUESTION 8 OF 9

Merge RELEASEDATE and LANGUAGE into a new RELEASEINFO column

TRANSFORMATION

Added a custom column combining Date.ToText([releasedate], “dd-MM-yyyy”) with the language field into a single “DD-MM-YYYY, Language” string.

Custom column — releaseinfo formula



SECTION · Currency Precision

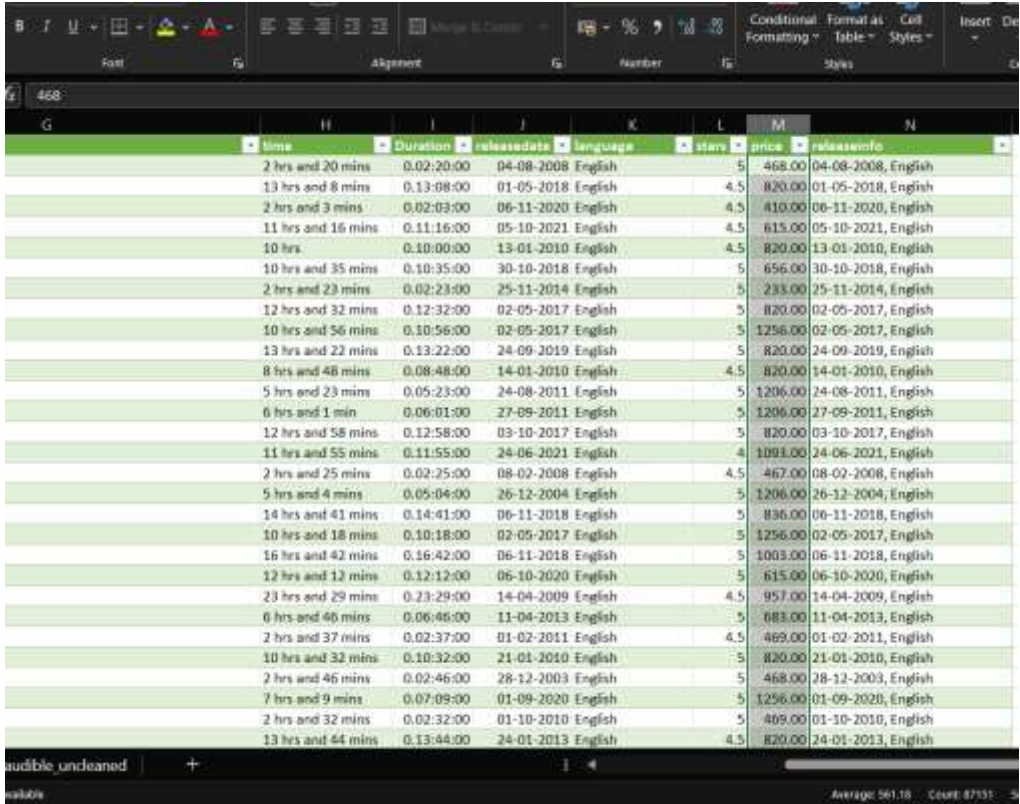
QUESTION 9 OF 9

Format all PRICE values with two decimal places

TRANSFORMATION

Standardized the numeric price column to display consistently with two decimal places, ensuring accurate and uniform currency formatting.

Result — price formatted to 2 decimals



	time	Duration	release date	language	stars	price	release info
	2 hrs and 20 mins	0.02:20:00	04-08-2008	English	5	468.00	04-08-2008, English
	13 hrs and 8 mins	0.13:08:00	01-05-2018	English	4.5	820.00	01-05-2018, English
	2 hrs and 3 mins	0.02:03:00	06-11-2020	English	4.5	810.00	06-11-2020, English
	11 hrs and 16 mins	0.11:16:00	05-10-2021	English	4.5	615.00	05-10-2021, English
	10 hrs	0.10:00:00	13-01-2010	English	4.5	820.00	13-01-2010, English
	10 hrs and 35 mins	0.10:35:00	30-10-2018	English	5	656.00	30-10-2018, English
	2 hrs and 23 mins	0.02:23:00	25-11-2014	English	5	233.00	25-11-2014, English
	12 hrs and 32 mins	0.12:32:00	02-05-2017	English	5	820.00	02-05-2017, English
	10 hrs and 56 mins	0.10:56:00	02-05-2017	English	5	1256.00	02-05-2017, English
	13 hrs and 22 mins	0.13:22:00	24-09-2019	English	5	820.00	24-09-2019, English
	8 hrs and 48 mins	0.08:48:00	14-01-2010	English	4.5	820.00	14-01-2010, English
	5 hrs and 23 mins	0.05:23:00	24-08-2011	English	5	1206.00	24-08-2011, English
	6 hrs and 1 min	0.06:01:00	27-09-2011	English	5	1206.00	27-09-2011, English
	12 hrs and 58 mins	0.12:58:00	03-10-2017	English	5	820.00	03-10-2017, English
	11 hrs and 55 mins	0.11:55:00	24-06-2021	English	4	1093.00	24-06-2021, English
	2 hrs and 25 mins	0.02:25:00	08-02-2008	English	4.5	467.00	08-02-2008, English
	5 hrs and 4 mins	0.05:04:00	26-12-2004	English	5	1206.00	26-12-2004, English
	14 hrs and 41 mins	0.14:41:00	06-11-2018	English	5	836.00	06-11-2018, English
	10 hrs and 18 mins	0.10:18:00	02-05-2017	English	5	1256.00	02-05-2017, English
	16 hrs and 42 mins	0.16:42:00	06-11-2018	English	5	1003.00	06-11-2018, English
	12 hrs and 12 mins	0.12:12:00	06-10-2020	English	5	615.00	06-10-2020, English
	23 hrs and 29 mins	0.23:29:00	14-04-2009	English	4.5	957.00	14-04-2009, English
	6 hrs and 46 mins	0.06:46:00	11-04-2013	English	5	683.00	11-04-2013, English
	2 hrs and 37 mins	0.02:37:00	01-02-2011	English	4.5	469.00	01-02-2011, English
	10 hrs and 32 mins	0.10:32:00	21-01-2010	English	5	820.00	21-01-2010, English
	2 hrs and 46 mins	0.02:46:00	28-12-2003	English	5	468.00	28-12-2003, English
	7 hrs and 9 mins	0.07:09:00	01-09-2020	English	5	1256.00	01-09-2020, English
	2 hrs and 32 mins	0.02:32:00	01-10-2010	English	5	409.00	01-10-2010, English
	13 hrs and 44 mins	0.13:44:00	24-01-2013	English	4.5	820.00	24-01-2013, English

What the Cleanup Delivered

1

Consistent Titles

Every book name now follows uniform Capitalize-Each-Word casing.

2

Split Contributors

Multi-author and multi-narrator fields separated into individual columns.

3

True Date & Duration Types

Release dates and listening times converted from text into native Excel types.

4

Clean Numeric Fields

Price and star-rating text converted to reliable decimal values, with errors resolved.

5

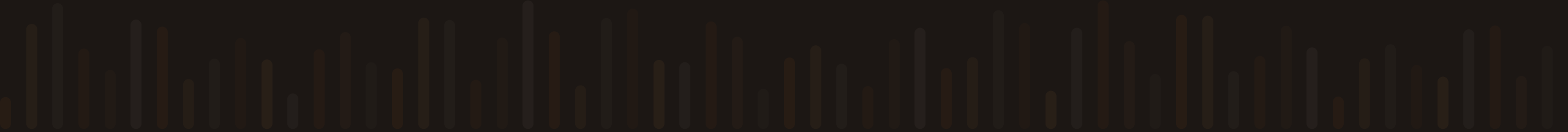
Merged Reporting Field

A new releaseinfo column combines date and language for faster reporting.

6

Precise Currency Formatting

All prices standardized to two decimal places throughout.



CONCLUSION

A Dataset Ready for Analysis

Using Power Query in Excel, the raw Audible export was transformed into a clean, consistent, and structured dataset. Standardized text fields, native date, duration, and numeric types, correctly split multi-value columns, and precise currency formatting mean the data can now be trusted for downstream analysis and visualization.

9

Case-study questions
answered

5

Transformation
categories covered

1

Analysis-ready
dataset produced

THANK YOU

Questions Welcome

Audible Data Cleaning Case Study — Power Query in Excel

Samarth Pathak

